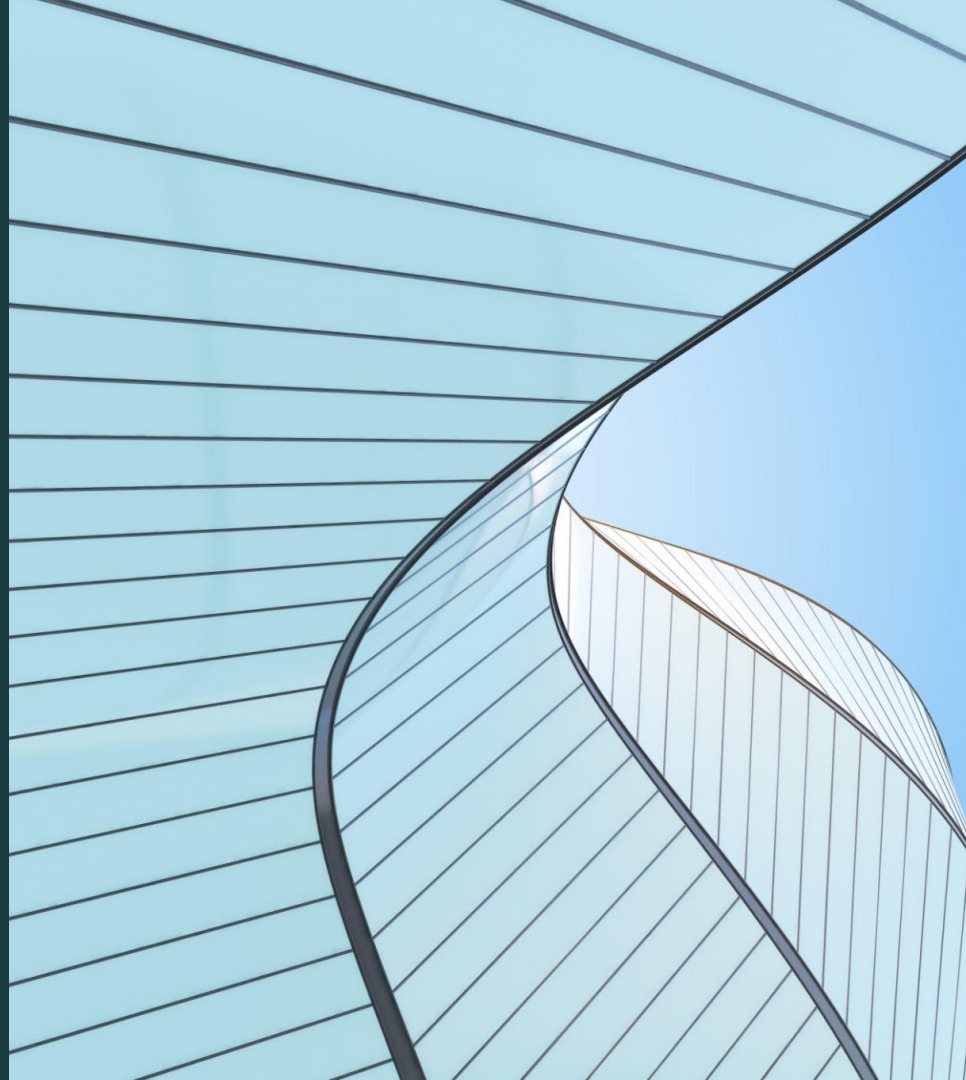
GR2

Daisy Zhong, Anvitha Kosuri, Lexuan Huang



Meet Our Team

Daisy:
Fourth year computer science major.

Anvitha Kosuri:
Fourth year computer science major.

Lexuan Huang:
Third year computer science major

Our Extra *Interviewees*

Game Shop Owner

Brian

- Non user of AI
- Owner of a cards, comic books, and games shop

Returning Full Time Student Nvidia

Andrea

- User of AI
- BS/MS computer science student conducting applied research, returning to NVIDIA

Brian

He is a non user of AI. He has used it before but not recently. He only uses AI to see the new advancements of AI and not so much in his day to day life

Why were they chosen/appropriate:

He was chosen because he gave diversity to our pool and could strengthen our data from our last non user in GR1.

Where interviews were conducted

They were conducted in his shop

How were they recruited

I went up to him while the shop was not busy to ask if I could interview him

Software used

Ipad for note taking and iphone to record the session

Brian

More Direct Quotes

- “These days, I don’t use AI at all, but I used to use it 2-3 times a week for image generating”
- “I feel more negative because it’s a tool that’s being used to create more capital instead of being used for good”
- “Those developing it are not moving it towards a positive potential, so now I see it as doing more harm than good”
- “I would like ai more if it was made more for a helpful tool and not used as a shortcut”
- “I would love to have AI give their sources and where they are drawing their information from”
- “Have AI not only show sources, but use past data to give the users how trustworthy the source is like show how the source could be wrong given xyz factors”
- “Humans should be in charge of the most of decision making when working with AI”

Surprises

- He doesn’t use AI, but he seems pretty knowledgeable about it(the effects of AI, how AI works…)
- As a whole he feels pretty negatively about AI, but it doesn’t seem like his negativity is directed AI more so as it’s directed at how people are using it. He thinks AI can be a positive and useful tool if used properly, but corporations are using it for their benefit instead

POV 1

We met Brian, surprised to notice that he feels negatively about AI yet still has hope for AI, wonder if this means if AI use were to be shift towards being more of a tool instead of shortcuts if he would feel more positively towards it, would be game changing to many people who view AI negatively

SAYS

- "I don't use AI anymore, even though I used it regularly before."
- "I would like AI more if it wasn't used as a shortcut."
- "AI should be designed as a genuinely helpful tool."
- "I want to know where AI information is coming from."
- "Those developing AI are not moving it toward positive potential."
- "Right now it feels like AI causes more harm than good."
- "I feel more negative toward AI now than before."
- "AI should explain why sources might be wrong."
- "AI is being used to generate capital instead of helping people."
- "I used AI two to three times a week mainly for image generation."

THINKS

- AI development is driven by profit rather than public good.
- Current incentives prevent ethical or human centered design.
- Human judgment should override automated suggestions.
- Trust should be earned, not assumed.
- Source visibility is essential for responsible AI use.
- Lack of transparency makes AI difficult to trust.
- AI should encourage learning, not shortcuts.
- AI credibility should be actively communicated.
- Users deserve context around information quality.
- AI systems hide too much uncertainty.
- Ethical design is currently undervalued.

DOES

- Previously used AI occasionally for creative generation.
- Avoids using AI tools entirely.
- Evaluates technology based on values, not convenience.
- Chooses not to engage with tools misaligned with beliefs.
- Avoids shortcuts even when available.
- Prefers evaluating information credibility through human judgment instead of accepting AI-generated conclusions at face value
- Distrusts AI outputs that lack citations or clear references, especially when information cannot be independently verified
- Seeks verified information manually by checking original sources rather than relying on AI-generated summaries
- Prioritizes human decision making in workflows.
- Reflects on who benefits from AI adoption

FEELS

- Disillusioned with current AI development.
- Frustrated by profit driven technological priorities.
- Concerned about misuse of powerful tools.
- Uncomfortable with opaque systems.
- Disappointed by missed positive potential.
- Emotionally distanced from AI systems.
- Protective of human autonomy.
- Distrustful of unverifiable information.
- Skeptical toward AI's societal impact.
- Cautious about re adopting AI tools.

SAYS

"I don't use AI anymore, even though I used it regularly before."

"I would like AI more if it wasn't used as a shortcut."

"AI should be designed as a genuinely helpful tool."

"I want to know where AI information is coming from."

"Those developing AI are not moving it toward positive potential."

Expresses strong concerns about AI's lack of transparency and repeatedly emphasizes the importance of sources, trust, and human control.

"AI us generate capital instead of helping people."

week mainly for image generation."

THINKS

AI development is driven by profit rather than public good.

Current incentives prevent ethical or human centered design.

Human judgment should override automated

Trust should be earned, not assumed.

Source visibility is essential for responsible AI use.

Believes AI development is driven by profit rather than public good and should prioritize accountability, credibility, and ethical responsibility.

AI credibility should be actively communicated.

information quality.

much uncertainty.

Ethical design is currently undervalued.

DOES

Previously used AI occasionally for creative generation.

Avoids

ooses not to gage with ls misaligned h beliefs.

Avoids shortcuts even when available.

Avoids relying on AI by manually verifying information, checking sources independently, and prioritizing human judgment over automation.

Prefers evaluating information credibility through human judgment instead of accepting AI-generated conclusions at face value

nually ginal than

Prioritizes human decision making in workflows.

information cannot be independently verified

AI-generated summaries

Reflects on who benefits from AI adoption

FEELS

Disillusioned with current AI development.

Frustra profit d techno prioritie

Disappointe d by missed positive potential.

Feels disillusioned and skeptical toward current AI systems, wanting greater transparency and reassurance before trusting AI support.

Emotionally distanced from AI systems.

Prote huma autonomy.

Cautious about re adopting AI tools.

Key Insights

Brian

Observation	Inference	Need
1. The participant frequently engages with AI systems in gaming contexts and interactive environments.	1.Playful and immersive use shapes expectations for AI to feel responsive and engaging.	1.Users want AI interactions that feel dynamic and adaptive without breaking immersion.
2. The participant values AI that reacts in real time and enhances gameplay experiences.	2. Responsiveness and feedback strongly influence perceived usefulness and enjoyment.	2. Users want AI that provides timely, context aware responses during active tasks..
3.The participant expresses interest in AI agents that behave like intelligent teammates or companions.	3.Social presence increases engagement but raises expectations for reliability and consistency.	3.Users want AI agents that feel collaborative while remaining predictable and controllable.
4.The participant prioritizes enjoyment and exploration over concerns about long term societal risks.	4. Immediate experiential value often outweighs abstract ethical considerations in entertainment contexts.	4. Users want AI systems that balance autonomy with boundaries appropriate for low stakes environments.

Key Insights Further Synthesis: Brian

Brian's interaction with AI is shaped primarily by experience quality rather than concerns about transparency or long term consequences. He values AI systems that feel responsive, engaging, and immersive, especially within low stakes environments such as gaming or interactive media. For him, the perception of intelligence comes less from an AI's ability to reason deeply and more from how naturally it reacts to user input in real time. Autonomy is acceptable when it enhances enjoyment and maintains flow, but becomes frustrating when it disrupts control or predictability. This suggests that in playful contexts, users are more willing to tolerate ambiguity and limited oversight as long as the interaction remains intuitive and enjoyable. Trust in these environments is built through consistent behavior and smooth feedback rather than explanations or visible reasoning, highlighting a contrast between entertainment driven use and productivity or decision focused contexts.

HMWs

For POV 1

How might we help people maintain agency and ownership when using AI?

How might we redesign AI to be more of a tool instead of shortcut

How might we influence corporations to use less AI

How might we encourage AI use that reinforces critical thinking rather than diminishing it?

How might we make AI more transparent so users can clearly see how it supports their skills

How might we figure out the appropriate amount to use AI

How might we add sources to AI when receiving answers

How might we build trust with AI-skeptical users by giving them control over how and when AI intervenes?

How might we reframe AI use to align with values(learning, creativity, and growth) other than efficiency

How might we rely on AI less and more on humans when making decisions

How might we make not a replacement for human effort or judgment?

Andrea

BS/MS student researcher returning full time to NVIDIA

Why were they chosen/appropriate:

They were chosen because their experience enables insight into agentic interaction, transparency, and human control.

Where interviews were conducted

They were conducted in the UCSB library.

How were they recruited

The participant was recruited through professional outreach on LinkedIn.

Software used

Laptop for note taking and iphone to record the session

Andrea

More Direct Quotes

- "I use AI every day for little things."
- "I ask it about health questions and random things about the world."
- "I use it for lecture clarification and practice problems."
- "I use AI a lot for coding syntax."
- "I also use it for brainstorming project ideas."
- "It enables people to learn a lot of new things more easily."
- "It's very accessible, which is a good thing."
- "People are beginning to rely on AI for really simple things."
- "It makes me sad because it feels like people are losing critical thinking skills."
- "There needs to be a middle ground."
- "Sketching, diagrams, and charts would feel the most natural."
- "Charts help with brainstorming and understanding concepts."
- "I definitely need to correct it sometimes."
  "I tell it what type of format I want it to reply in."
- "Having the option to strictly control the scope of what AI is helping you on would be nice."

Surprises

- Despite appreciating AI's accessibility and learning benefits, she feels emotional discomfort when it replaces simple human thinking.
- Although she is technically experienced, her concerns about AI are rooted more in values and responsibility than usability.
- Rather than wanting more autonomous features, she prefers systems that slow her down and make reasoning visible.
- While AI is marketed as saving time, Andrea values understanding and control over speed or convenience.

POV 1

We met Andrea, and were surprised to notice that although she uses AI every day, she remains deeply cautious about its growing influence and loss of human critical thinking. This made us wonder whether shifting AI toward being more transparent, steerable, and supportive rather than autonomous could help users feel more confident and in control. Such a shift could be game changing for people who rely on AI but struggle with trusting its long-term impact.

SAYS

"Visual explanations make complex concepts easier to process."

"I like when AI explains its reasoning step by step."

"I mostly use it for search and school help."

"People are starting to rely on it too much."

"Voice interaction could be useful, just not something I personally default to."

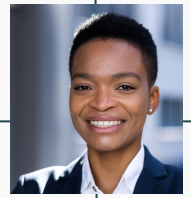
"I use AI every day for small things."

"Voice interaction could be useful, just not something I personally default to."

"Typing still feels like the most natural way to interact with AI."

"Charts and diagrams help me understand ideas much faster."

"People are already becoming too dependent on AI for simple thinking."



THINKS

AI adoption is accelerating faster than society has time to reflect.

Over Reliance on AI feels inevitable without better design boundaries.

Designers may prioritize innovation over long-term responsibility.

Users need clearer ways to control an AI's scope of action.

Transparency in reasoning matters more than flashy interaction modes.

AI should enhance learning rather than replace independent thinking.

Text explanations feel more trustworthy than abstract summaries.

Many futuristic interaction modes sound interesting but lack real payoff.

Visual reasoning can support understanding when used intentionally.

AI uncertainty should be visible before users rely on outputs.

DOES

Uses AI daily for search, clarification, and general academic support.

Brainstorms project ideas using AI assistance.

Relies on AI for coding syntax and debugging help.

Asks AI to explain lecture material in simpler terms.

Uses AI to practice problems and understand concepts.

Corrects AI responses when they do not meet expectations.

Specifies preferred formats and response styles explicitly.

Mentally checks AI outputs before accepting them.

Adjusts trust based on task importance and risk.

FEELS

Uneasy about long term societal and economic consequences.

Cautiously engaged rather than overly excited.

Calm but vigilant while interacting with AI systems.

Comfortable using AI as support, not authority.

Conflicted about AI's growing role in everyday life.

Concerned about diminishing human critical thinking.

Uncomfortable when AI appears too independent.

Skeptical of hype around futuristic agentic designs.

Relieved when AI behavior is predictable and transparent.

SAYS

"Visual explanations make complex concepts easier to process."

"I like when AI explains its reasoning

"I mostly use it for search and school help."

"People are starting to rely on it too much."

"Voice interaction could be useful, just not something I personally default to."

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"Ch diagr me understand ideas much faster."

too dependent on AI for simple thinking."

Uses AI frequently for academic and everyday tasks, but repeatedly emphasizes the importance of staying in control and avoiding overdependence.

THINKS

AI adoption is accelerating faster than society has time to reflect.

Over Reliance on AI feels inevitable without better design

Designers may prioritize innovation over long-term

Users need clearer ways to control an AI's scope of action.

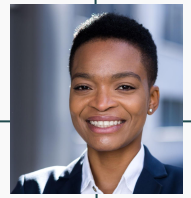
Transparency in reasoning matters more than flashy interaction modes.

Many futuristic interaction modes sound interesting but lack real payoff.

understanding when used intentionally.

nty visible before users rely on outputs.

Believes AI should remain a transparent, efficient support tool that enhances human thinking without shaping behavior or making decisions autonomously.



DOES

Uses AI daily for search, clarification, and general academic support.

Asks AI to xplain lecture material in mpler terms.

Uses AI to practice problems and understand concepts.

Corrects AI responses when they do not meet expectations.

s AI

Adjusts trust based on task importance and risk.

Uses AI selectively and actively monitors its outputs, correcting responses and adjusting trust to maintain accuracy and personal agency.

FEELS

Uneasy about long term societal and economic consequences.

Calm but

Comfortable using AI as rt, not ity.

Conflicted about AI's growing role in everyday life.

al of ound

Relieved when AI behavior is predictable and transparent.

Feels comfortable receiving AI assistance, but uneasy when systems act autonomously or adapt behavior without clear visibility.

Key Insights

Andrea

Observation	Inference	Need
1. The participant previously used AI moderately but has since stopped using it entirely.	1.Ethical concerns can override convenience even when AI is useful.	1.Users want AI tools that align with personal values and ethical expectations.
2. The participant believes AI is currently driven by capital generation rather than social good.	2. Perceived developer intent significantly impacts trust in AI systems.	2. Users want AI development that prioritizes societal benefit over profit incentives.
3. The participant strongly emphasizes the importance of source citation and credibility.	3. Lack of transparency prevents users from trusting AI generated information.	3. Users want AI to clearly show sources and communicate information reliability.
4. The participant insists that humans should remain responsible for major decisions.	4. Autonomous AI behavior creates discomfort when accountability is unclear.	4. Users want AI systems that defer to human judgment and maintain clear decision boundaries.

Key Insights Further Synthesis: Andrea

Andrea's relationship with AI is defined by frequent use paired with persistent caution and self monitoring. While she relies on AI daily for academic support, clarification, and brainstorming, she remains deeply concerned about over dependence and the gradual loss of critical thinking skills. This creates an ongoing tension between convenience and responsibility, where usefulness does not automatically translate into trust. Andrea values AI as a supportive tool rather than an autonomous agent and consistently emphasizes the importance of transparency, steerability, and clearly defined boundaries. Her preference for visual explanations, diagrams, and visible reasoning reflects a desire to understand how conclusions are formed rather than simply receiving answers. For Andrea, trust emerges not from automation but from legibility, where the AI's behavior, uncertainty, and scope of action are made visible so that human judgment remains central throughout the interaction.

HMWs
For POV 2

How might we design AI that encourages understanding rather than simply delivering answers?

How might we help users actively participate in an AI's reasoning process?

How might we make AI uncertainty visible before users rely on its outputs?

How might we allow users to define clear boundaries for what AI can and cannot do?

How might we encourage moments of reflection within AI interactions without slowing users down unnecessarily?

How might we make AI reasoning legible without overwhelming users with information?

How might we design AI that supports learning without weakening critical thinking skills?

How might we help users feel confident that AI is not shaping decisions without consent?

How might we create AI systems that reinforce human judgment rather than replace it?

How might we design AI that adapts without feeling intrusive or unpredictable?

How might we help users stay reflective rather than passive while using AI?

Top 3 HMW's

1. **How might we redesign AI to act as a supportive tool rather than a shortcut for thinking?**
2. **How might we make AI reasoning and uncertainty visible so users can better judge when to trust it?**
3. **How might we help users maintain agency and control while working alongside increasingly capable AI systems?**

Solutions for HMW #1

How might we redesign AI to act as a supportive tool rather than a shortcut for thinking?

1. Provide structured prompts that guide users without revealing final answers
2. Introduce reflection checkpoints before revealing solutions
3. Encourage users to explain their reasoning during problem solving
4. Offer partial hints that unlock progressively
5. Frame responses as learning pathways instead of conclusions
6. Visualize concept relationships rather than outcomes
7. Require minimal user input before advancing assistance
8. Present multiple approaches without prioritizing one
9. Turn explanations into interactive walkthroughs
10. Highlight what the user already understands
11. Support exploration through examples instead of direct answers
12. Delay full responses unless explicitly requested
13. Encourage curiosity through follow-up questions
14. Reframe assistance as coaching rather than completion
15. Adapt support level based on engagement behavior

Solutions for HMW #2

How might we make AI reasoning and uncertainty visible so users can better judge when to trust it?

1. Display confidence indicators alongside responses
2. Separate reasoning from final output visually
3. Highlight assumptions used to reach conclusions
4. Show alternative interpretations side by side
5. Indicate where information may be incomplete
6. Use visual cues to represent uncertainty levels
7. Summarize source reliability contextually
8. Flag speculative content distinctly from factual claims
9. Provide explanations for why an answer could be wrong
10. Allow users to expand or collapse reasoning layers
11. Visualize missing or ambiguous data
12. Compare multiple reasoning paths transparently
13. Surface confidence changes when new input is added
14. Offer credibility summaries instead of raw citations
15. Prompt user review when certainty is low

Solutions for HMW #3

How might we help users maintain agency and control while working alongside increasingly capable AI systems?

1. Allow users to define task boundaries at the start
2. Introduce checkpoints before major actions occur
3. Enable adjustable levels of system involvement
4. Require confirmation at key decision moments
5. Provide clear previews of intended actions
6. Support easy undo and rollback options
7. Let users pause or override system behavior
8. Offer transparency dashboards showing system activity
9. Enable consent based adaptation over time
10. Give users ownership of personalization settings
11. Make system behavior reviewable after completion
12. Allow autonomy toggles during active tasks
13. Encourage collaborative turn taking between human and system
14. Keep humans responsible for final decisions
15. Default interactions to observation rather than execution

Top 3 solutions

1) **Introduce reflection checkpoints before revealing solutions**

   AI requires user reflection or partial reasoning before revealing solutions, encouraging learning instead of shortcutting.

2) **Separate reasoning from final output visually**

   AI visually separates reasoning from final answers so users can evaluate how and why conclusions are made

3) **Enable adjustable levels of system involvement**

   Users control how much the AI can intervene, maintaining agency and preventing over-reliance.

Experience Prototype

Solution 1: Introduce reflection checkpoints before revealing solutions

Critical Assumption: Users will not feel frustrated when AI delays answers and asks them to reflect first.

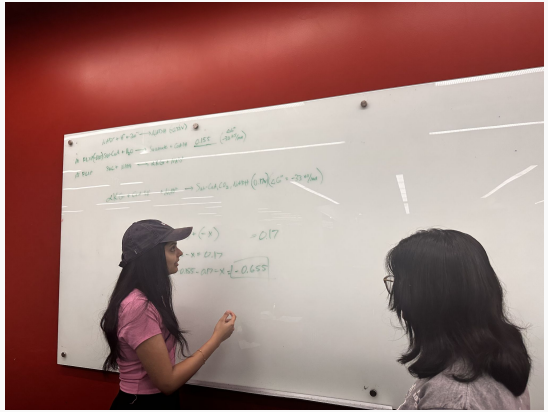
Scene: A student works on a short problem with another person acting as a tutor who intentionally delays giving answers.

Props: Whiteboard, Whiteboard marker

How it was run

The participant solved a problem while the tutor intentionally paused and required reflection before giving hints or answers. This directly tested whether delaying answers would frustrate the user or help them feel more engaged.

Experience Prototype



Participants:

1) A freshman studying Chem 1A
2) A senior year who took 1A previously tutoring her

How were they recruited?

- Both participants were recruited through personal connections in chemistry classes and tutoring spaces on campus.
  They were asked in person if they would participate in a short study-related activity.

Why are they relevant?

- The freshman represents a user who needs clear explanations and support while learning new material.
- The senior year student represents a more experienced user who cares about understanding reasoning and evaluating correctness.
- Together, they reflect different levels of math confidence and different needs for trust and transparency when using AI for problem solving.

Experience Prototype

Script

1. The tutor gives the participant a math problem and says:
   "Try solving this on your own. I'll help, but I won't give the answer right away."

2. After the participant struggles or asks for help, the tutor pauses and asks:
   - "What do you think the first step should be?"
   - "What have you tried so far?"
   - "Why do you think that approach might work?"

3. The participant explains their thinking aloud and writes a short response.

4. Only after reflection, the tutor provides a small hint (not the full answer).

5. Steps 2–4 repeat once more before the final solution is shown.

6. At the end, the participant fills out a short survey rating:
   Frustration level
   - Engagement
   - Whether the reflection helped understanding

Result

What worked

- The freshman became more engaged when asked to explain their thinking before receiving help.
- The tutor found that pausing for reflection revealed where the student was actually confused.
- The interaction felt more like collaborative problem solving rather than answer giving.
- Both participants reported that reflection made the solution feel more meaningful once revealed.

What didn’t

- The freshman showed slight frustration when stuck and wanting the answer quickly.
- Reflection pauses felt awkward when the student had no idea how to proceed.
- The tutor had to carefully judge when to intervene to avoid slowing progress too much.

Result

New learnings:

- Reflection checkpoints are helpful when users have partial understanding, but not when they are completely lost.
- The timing of the pause matters more than the pause itself.
- Guidance questions work better than silence during reflection moments.

Was the assumption valid?

- **Partially valid.** The student was not overly frustrated, but reflection was only helpful when paired with gentle prompts rather than full delay.

What does this mean going forward?

- Future designs should include optional or guided reflection rather than forcing long delays.
- Reflection should adapt based on user confidence level.
- AI should encourage thinking without blocking progress entirely.

Experience Prototype

Solution 2: Separate reasoning from final output visually

Critical Assumption: Users will feel more confident deciding what to trust when uncertainty and reasoning are visible instead of hidden.

Scene: A participant is asked to make a small real-life decision using information provided by another person.

Props: One printed question, two printed answer cards, short survey

How it was run

The participant was shown two responses to the same question: one with only a direct answer and one with reasoning and a short statement of uncertainty. They were asked to choose which response they trusted more and explain why. This allowed us to directly observe whether visible reasoning and uncertainty influenced their trust and decision-making.

Experience Prototype

Solution 2: Separate reasoning from final output visually

Participants:

1) UCSB student majoring in Biology. Frequently ask health-related questions.
2) UCSB student majoring in Computer Science. Uses AI regularly and is interested in understanding how answers are generated.

How were they Recruited?
Both participants were recruited through personal outreach to classmates who use AI in their daily academic work and had no prior involvement with the project.

Why are they relevant?

1. The Biology student represents users who rely on AI for real-world decisions where trust and uncertainty matter, such as health and studying.

2. The Computer Science student represents technically engaged users who care about how reasoning is presented and want transparency when using AI tools.

Experience Prototype

Script

Step 1 – Present the task:
The researcher gives the participant the printed question and says:
“Please read this question as if you were trying to make a real decision.”

Step 2 – Show responses:
The researcher shows the participant the first answer card (direct answer only), then shows the second answer card (answer with reasoning and uncertainty).

Step 3 – Choice:
The researcher asks:
“Which of these two responses do you trust more?”

Step 4 – Explanation:
The participant explains why they chose that response and what made it feel more or less trustworthy.

Step 5 – Survey:
The participant completes a short survey about their confidence and trust in each response.

Result

What worked

- Participants felt more confident when the reasoning was separated from the final answer.
- Both users said the structure made the answer feel more transparent and trustworthy.
- The biology student appreciated being able to check the logic before trusting the result.
- The CS student liked seeing how conclusions were reached instead of just the outcome.

What didn't

- The reasoning section was sometimes ignored when users were in a hurry.
- One participant found the explanation distracting for simple questions.
- The layout felt slightly overwhelming when both sections were visible at the same time.

Result

New learnings:

Users want access to reasoning, but only when they choose to view it. Making reasoning optional or collapsible may be more effective than always showing it.

Was the assumption valid?
Partially validated. Separating reasoning from the final answer improved trust, but only when the reasoning was short and clearly structured.

What this means going forward:
Our design should visually separate answers and reasoning while allowing users to control how much explanation they see. This supports transparency without overwhelming users, helping maintain trust and usability at the same time.

Experience Prototype

Solution 3: Enable adjustable levels of system involvement

Critical Assumption: Users will feel more in control and less dependent when they can choose how much help they receive.

Scene: A student works on a CS problem with another person acting as a tutor who can provide different levels of help based on what the student requests.

Props: Printed problem sheet, three labeled cards (“No Help”, “Hint Only”, “Full Explanation”), notebook and pen, short Google Form survey.

How it was run

The participant solved a problem while choosing between three levels of help: no help, a small hint, or a full explanation.

This directly tested whether having control over help level affected confidence and reliance.

Experience Prototype

Solution 3: Enable adjustable levels of system involvement

Participants:

1) A sophomore year CS student studying for Data Structures midterm
2) A junior-year CS student tutoring

How were they Recruited?

- Recruited through personal connections in CS classes and tutoring spaces on campus.
- Asked in person to participate in a short study-related activity..

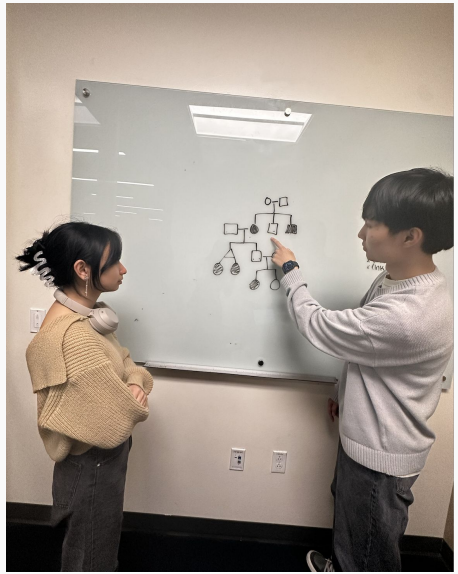
Why are they relevant?

1. The freshman represents users who may over-rely on help when stressed.
2. The tutor represents experienced users who value maintaining problem-solving agency.
3. Together, they reflect different needs for control and support when using AI.

Experience Prototype

Script

1. **Setup:**
   Facilitator explains: “You will solve this problem. When you want help, choose one of these cards to decide how much help you receive.”



2. **Task:**
   Participant works on the problem and selects a card when stuck.
   Facilitator responds based on the card:

- **No Help:** Encourages independent thinking.
- **Hint Only:** Gives a small directional hint.
- **Full Explanation:** Explains the next step.

3. **Wrap-Up:**
   Participant completes the problem and answers a short survey about confidence, frustration, and reliance on help.

Result

What worked

- The freshman felt more confident when choosing to receive only hints instead of full solutions.
- The tutor observed more independent thinking when the student controlled help level.
- Both participants said the system felt more respectful of their ability.

What didn’t

- The freshman sometimes chose full explanations too quickly when anxious.
- Switching between help levels interrupted problem flow slightly.
- The student needed guidance on when to use each level.

Result

New learnings:

- Users like having control, but still need encouragement to avoid defaulting to maximum help.
- Simple help levels are easier to use than complex settings.
- Agency increases when assistance feels optional rather than automatic.

Was the assumption valid?
Mostly valid. Users felt more in control, but still gravitated toward high assistance under stress.

What this means going forward:

- AI systems should offer adjustable help with gentle nudges toward independence.
- Defaults should favor low intervention with escalation available.
- Design should focus on making control simple and visible.

Final (Combined) Solution

We will move forward with a combination of:

Reflection checkpoints before revealing solutions and **adjustable levels of system involvement**.

Together, these support learning without forcing users into either full dependence or full independence, and allow assistance to adapt to different confidence levels and situations.

Final (Combined) Solution

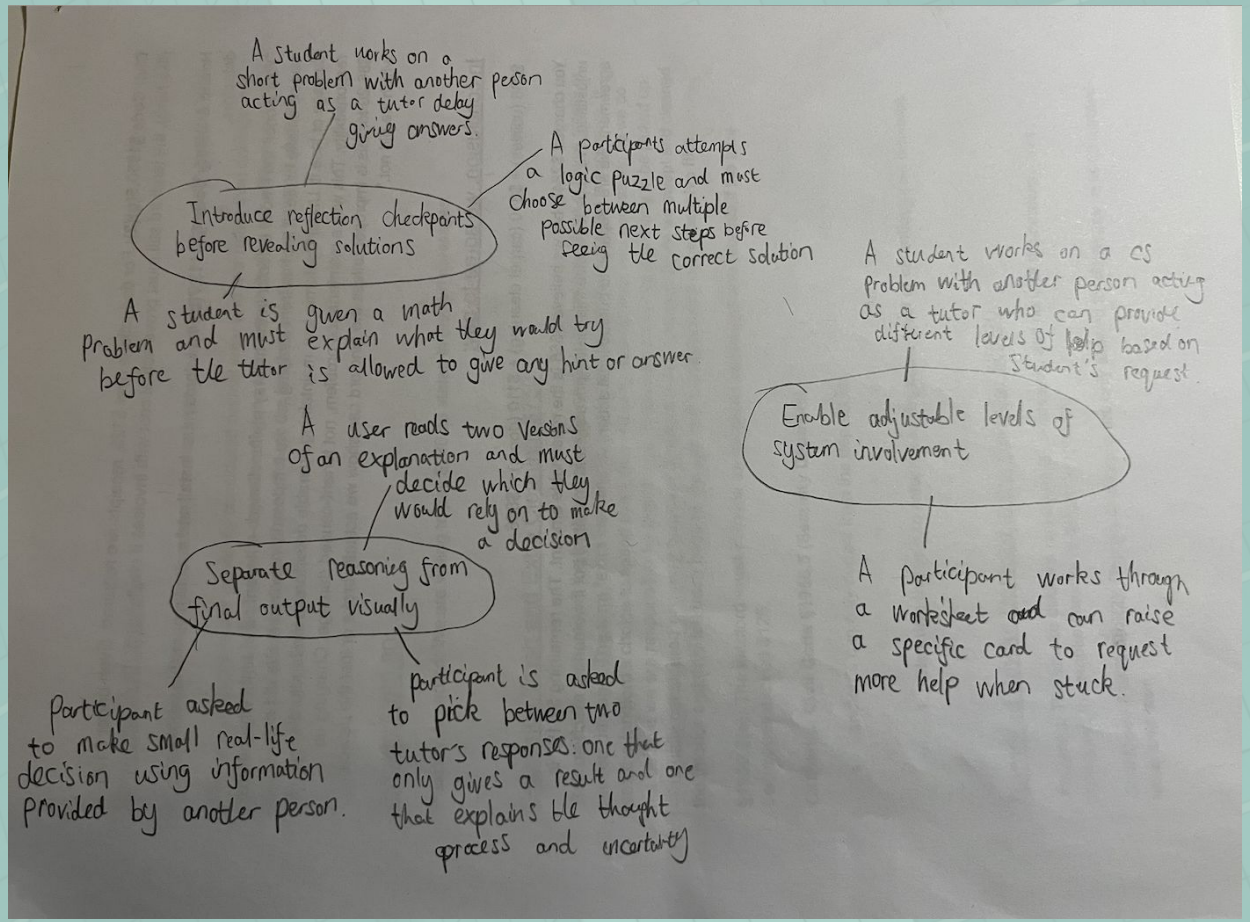
Who does it serve?

This solution serves students and learners who want to understand their work rather than just receive answers, especially in subjects like math or mention technical problem-solving. It is useful for users who value having control over how much help they get and who are open to reflecting on their thinking.

Who might it leave out?

It may leave out users who want instant answers with no extra steps or interaction. It may also be less suitable for users who are very time-constrained or who feel discouraged by reflection prompts instead of direct solutions.

Appendix Brainstorming Pictures



Appendix

1. **Participants**
- Brian — Game shop owner, limited AI use, values-driven perspective
- Andrea — BS/MS Computer Science student, daily AI user with concerns about over-reliance

2. **Interview Context**
- Interviews conducted in participant-relevant environments (shop, UCSB library)
- Data collected through direct quotes, observation, and follow-up discussion

3. **Research Artifacts**
- Empathy maps synthesizing what users say, do, think, and feel
- Key insight tables mapping observations → inferences → user needs

4. **Key Themes Identified**
- Preference for AI as a supportive tool, not a shortcut
- Importance of visible reasoning and uncertainty
- Desire to maintain human agency and control
- Concerns around loss of critical thinking and over-dependence

5. Design Translation

- Insights informed POV statements and “How Might We” questions
- Top HMWs focused on learning support, transparency, and user agency

6. Evaluation Method

- Low-fidelity experience prototypes used to test assumptions
- Prototypes examined reflection timing, reasoning visibility, and adjustable assistance
- Findings informed final combined solution direction